



BRAC University
Department of Computer Science and Engineering
CSE 443: Bioinformatics - I
Quiz 01: Fall 2025 Time: 15 Minutes Marks: 10

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1. GibbsSampler is guaranteed to find the global optimum motif. (True/False)
2. Okazaki fragments are synthesized only on the lagging strand. (True/False)
3. A (500,3)-clump of a 9-mer means it appears $\geq 3\times$ in every 500-bp window. (True/False)
4. The most probable 3-mer from the profile $A=[0.6, 0.1, 0.3]$, $T=[0.2, 0.7, 0.1]$, $G=[0.1, 0.1, 0.5]$, $C=[0.1, 0.1, 0.1]$ is: (A) ATG (B) AGG (C) TTG (D) ACG
5. DNA polymerase can initiate synthesis without a primer. (True/False)
6. Minimum skew value is always negative. (True/False)
7. Entropy of column (0,0,0.9,0.1) is closest to:
A. 0.33 B. 0.47 C. 0.64 D. 0.91
8. Gibbs vs RandomizedMotifSearch: key difference is:
A. uses EM B. discards one sequence per iteration C. deterministic D. needs phylogeny
9. GC-skew sign at terC is opposite to that at oriC:
A. true B. false C. species-specific D. undefined
10. Laplacian correction turns zero into:
A. $1/(t+4)$ B. $1/t$ C. $1/(t+1)$ D. $1/4$